

Objective

Mechanical engineering student seeking a summer internship focused on sensor systems, real-time control, and mechanical-aerodynamic design. Interested in applying CAD, simulation, signal processing, and modeling to experimental hardware and sensing technologies.

Education

Johns Hopkins University, Whiting School of Engineering

Baltimore, MD

B.S. in Mechanical Engineering

Anticipated May 2029

Relevant Coursework: CAD and Design, Mechanics, Python, Digital Electronics, Electricity and Magnetism, Multivariable Calculus

Technical Skills

- **Software:** SolidWorks, ANSYS (Structural, Fluent, CFX), MATLAB, ROS 2, Webots, OnShape, Multisim
- **Programming/Analysis:** Python, MATLAB, C++, FEA, CFD, FFT, control systems, data visualization
- **Prototyping:** 3D printing, laser cutting, soldering, Arduino, mechanical assembly and testing

Engineering Experience

Amarkosha Wind Harvester Start-Up

Baltimore, MD

Mechanical Lead, Product Development

Jul 2025–Present

- Led **VIV wind harvester** design and multiphysics simulation in SolidWorks and ANSYS.
- Performed two-way transient **FSI analysis** (Fluent/CFX + Structural) to assess aerodynamic-structural coupling.
- Applied **FFT-based signal processing** in MATLAB to extract vortex shedding and lock-in behavior.
- Optimized geometry for wake effects, yielding **166% wind-speed amplification**.
- Accepted to **SPARK Accelerator Program** and selected as **BII Grant finalist**.

Johns Hopkins Mars Rover Team – Mechanical Subteam

Baltimore, MD

Mechanical & Controls Member (Wrist Mechanism)

Sep 2025–Present

- Designed a **3-DOF wrist** to enhance manipulator dexterity and payload handling.
- Integrated **ROS 2** for real-time control and sensor feedback.
- Conducted **torque analysis** for motor and joint optimization.

GreenWorks Development

Mechanicsburg, PA

Technical Sales & Project Intern

Jan–Aug 2025

- Modeled solar systems in Helioscope and Excel for a \$5M performance validation project, contributing to proposal presentations with corporate leadership.

UPMC Magee–Women’s Hospital

Pittsburgh, PA

Research Assistant, Dept. of Anesthesiology

Jun–Sep 2024

- Processed and visualized clinical datasets in R to support pre-eclampsia diagnostics research.

Projects

4-DOF Robotic Arm Simulation (Independent)

- Designed and simulated a 4-DOF manipulator in **SolidWorks** and **Webots** with **ROS 2** control.
- Integrated Arduino actuation and inverse kinematic feedback for closed-loop motion.
- Applied **Design for Manufacturing (DFM)** principles and **static FEA** to ensure mechanical reliability.

Leadership & Honors

- **President & Founder, Renewable Energy Club (RENEW)** — Led sustainable energy design projects and guest seminars.
- **Facilities Chair, Mini-THON** — Managed logistics and volunteers for \$280K pediatric cancer fundraiser.
- **Honors:** Science Hall of Fame, William R. Pierce Physics Award, Lois Wolf GPA Athlete Award, Melissa Huang Scholarship.